

Unit 2: Implementation

This unit introduces you to the creation of dynamic routing channel template. It helps to determine consumers dynamically during transaction processing, handle processing of various file layers, and to generate reports. You will also learn extensibility of Sterling File Gateway user interfaces, creation of custom protocols and event codes.

LESSON 2.1: Dynamic Route

What this lesson is about

This lesson describes about the Sterling File Gateway dynamic routing channel template, broadcast messages, mass redelivery, and mass replay.

What you should be able to do

- Describe the major functionality of dynamic routing channel.
- Configure a dynamic routing channel template.
- Apply business process to determine consumers.
- Explain and configure broadcast messages using dynamic routing channel template.
- Explain mass replay and redelivery of processed transaction.

How you will check your progress

The progress of the course is analyzed based on the successful applications of the hands-on exercises.

Additional References

Documentation

- Broadcast routing message:
http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg222_updates.doc/sfgcreatebroadcastrct.html?lang=en
-

Dynamic Routing

Introduction

Dynamic route describes the capability of a system through which the routes are characterized by their destination. A dynamic routing channel template determines the consumer at the time of file transfer.

A dynamic routing channel template specifies one of the following listed methods to determine the consumer for each file transfer:

- Specify the ConsumerName or ConsumerCode system facts in the producer file structure.
 - Such as (.+)_(.+)[.](.+) The characters “.” indicates consumerName, the preceding underscore in the file name determines the routing of the file to a consumer. The consumer is identified in the outer layer or the first layer of an outer zip layer.
- Specify a business process and ProcessData element name
 - create and deploy a business process that returns an XML element, ProcessData. The Process Data elements contains the consumer name.

Editing routing channel template

Routing Channel Templates are edited when there is no routing channels associated with them. When a Routing Channel Template is selected for editing, the following components available for modification:

- Type
 - Special Characters
 - Groups
 - Provisioning Facts
 - Producer
 - Consumer
-

Exercise 2.1.1: Create dynamic outbound route

Introduction

In this exercise, you create a routing channel template and routing channel to move data dynamically. However, the consumer's mailbox can be derived at runtime, using part of the file name. File name dictates which mailbox receives the data and compresses the data in zip format for the consumer:.

Parameter	Value
Producers Input Data	partner_NewFile.txt
Producers mailbox	/abcde/
Consumers Output Data	NewFile.zip
Consumers mailbox	/partner/inbox

This exercise consists of the following four steps:

1. Create a routing channel template.
2. Create route channel.
3. Upload input data file.
4. Analyze partner myfilegateway interface.

Instructions

Complete the following steps to create a routing channel template:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the main menu, select **Routes > Templates**.

Step 3: Click **+Create**.

Step 4: Enter template Name: **Dynamic_outbound**.

Step 5: Enter template Type: **Dynamic**.

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Exercise 2.1.1: Create dynamic outbound route

(Continued)

Instructions

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Step 6: Select, **Facts in producers file structure** to determine the consumer from facts of the producers file structure.

Step 7: From the **Special Characters** tab, select, **None** for the **Special Characters**.

Step 8: From the **Groups** tab, click, **+Add** for the **Producer Groups**.

Step 9: From the drop down list, select **All Partners**.

Step 10: Click **+Add** to add the **Consumers Groups**.

Step 11: From the drop down list, select **All Partners**.

Step 12: From the **Provisioning Facts** tab, accept the default values for the **Provisioning Facts**.

Step 13: From the **Producer** tab, click **+Add** to add the **Producer**.

Step 14: From the **Producer File Structure** screen, specify the following values:

Parameter	Value
Producer File Type	Text
File name pattern as regular expression	(.+)_(.+).txt
File name pattern group fact names, comma delimited	ConsumerName,myFilename

Step 15: Click **Save**.

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Exercise 2.1.1: Create dynamic outbound route

(Continued)

Instructions

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Step 16: From the **Consumer** tab, click **+Add** to add the **Consumer**.

Step 17: Accept the default values for mailbox path.

Step 18: Select **If checked, mailboxes matching this pattern may be created on demand**.

Step 19: Click **+Add** to access the **Consumer File Structure**. Specify the following values:

Parameter	Value
Consumer File Type	Zip
File name format	\${myFilename}.zip

Step 20: Click **Next**.

Step 21: In the **Consumer File Structure** window, specify the following values:

Parameter	Value
Consumer File Type	Text
File name format	\${myfilename}.txt

Step 22: Click **Save** three times. The message, Routing Channel Template successfully created, is displayed, and Click **OK**.

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Exercise 2.1.1: Create dynamic outbound route

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Instructions

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Complete the following steps to create route channel:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the main menu, select **Routes > Channels**.

Step 3: Click **+Create**.

Step 4: From the **Create a New Routing Channel**, specify the following values:

Parameter	Value
Routing Channel Template	Dynamic_outbound
Producer	abcde

Step 5: Click **Save**. The message, Successfully created Routing Channel, is displayed.

Step 6: Click **OK** and log out filegateway interface.

Complete the following steps to upload input data file:

Step 1: Log in to **myfilegateway** by specifying **abcde** as the user ID and **password** as password.

Step 2: Click the **Upload Files** tab.

Step 3: From the Mailbox path section, select, / [forward slash].

Step 4: From the **File** menu, select **C:\6F86G\ClassFiles\partner_NewFile.txt**.

Step 5: Click **Send**. The file is routed to /partner/inbox.

Complete the following steps to analyze partner myfilegateway:

Step 6: Log in to myfilegateway with user ID as **partner**, password as **password**.



Note: _____
If the partner screen is not updated, refresh your browser.

Step 7: Select the **File Activity** tab.

Step 8: Click **Find** button in Search Criteria panel to view the arrived files in partner's inbox.

Step 9: Click **Logout**.

Exercise 2.1.2: Build mailbox add business process

Introduction

In this exercise, you create a business process to insert a file into the mailbox repository through a business process. This way the *abc* company will not be required to log in into myfilegateway to start the file transfer. The business process inserts the file into the producer's mailbox and Sterling File Gateway routes the data to the consumer's mailbox. The key to making the exercise work is to execute the business process with the producer's user ID.

The exercise is a four step process:

1. Build the mailbox to add a business process.
2. Check in the business process.
3. Execute the business process.
4. Analyze the result.

Instructions

Build the mailbox to add business process

Complete the following steps to build the mailbox to add business process:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Select **Tools > B2B Console**.

Step 3: From the home page, click **Go**, to open the Graphical Process Model (GPM).

Step 4: From the GPM tool, click **File > New**.

Step 5: Click **View > Stencil > BPML**.

Step 6: Click **View > Stencil > All Services**.

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Exercise 2.1.2: Build mailbox add business process

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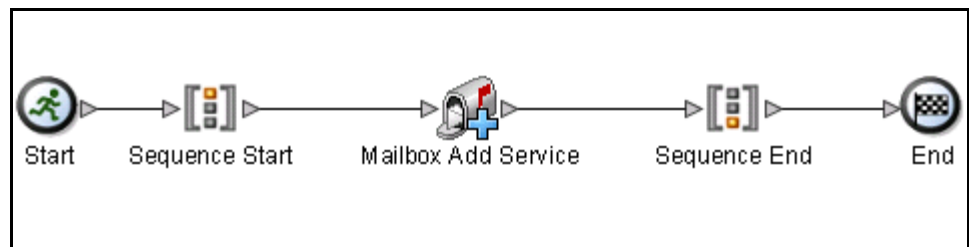
Instructions

....(Continued)

Step 7: Move the following stencils to your workspace.

- Start (1)
- End (1)
- Sequence Start (1)
- Sequence End (1)
- Mailbox Add Service (1)

Step 8: Arrange and connect the stencils as shown in the following screen shot:



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Exercise 2.1.2: Build mailbox add business process

(Continued)

Instructions

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Step 9: Double-click the Mailbox Add Service. In the Service Editor – Mailbox Add Service, specify the following values:

Name	Value
Config	MailboxAdd
ExtractableCount	1
MailboxPath	/

Step 10: Navigate to, **File > Save As**. Validate the business process and name the business process as **6F86Gmailbox_add.bp**.

Check in the business process

Complete the following steps to check in the business process:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: Click **Business Process > Manager**.

Step 4: From the **Create** section, next to **Process Definition**, click **Go!**

Step 5: From the Editor: Process Name dialog box, specify the following values:

Parameter	Value
Name	6F86Gmailbox_add
Select an input mode for defining the new process	Check in business process created by the graphical modeling tools.

Step 6: Click **Next**.

Step 7: From the Editor: Process: **6F86Gmailbox**: Check-in dialog box, click **Browse** and select **6F86Gmailbox_add.bp**. Click **Open**.

Step 8: From the Editor: Process: **6F86Gmailbox**: Check-in dialog box, in the **Description** field, type **Mailbox file add process** and click **Next**.

Step 9: Accept the default values for the remaining parameters and click **Next**. Verify that the Business Process is enabled when you click **Finish**.

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Exercise 2.1.2: Build mailbox add business process

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Instructions

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Execute the business process

Complete the following steps to execute the business process:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: From the B2B Console, click **Business Process > Manager**.

Step 4: Search for **6F86Gmailbox_add**.

Step 5: From the search result screen, click **execution manager**.

Step 6: From the **execution manager** screen click **Execute**.

Step 7: Click, **Browse**. Access **partner_NewFile.txt** from **C:\6F86G\Classfiles** folder.



Note: _____

Make sure the file name begins with partner name, as specified in the routing channel template facts.

Step 8: From Execute Business Process Input screen, specify **abcde** in the Run As User parameter and click **Go!**.

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Exercise 2.1.2: Build mailbox add business process

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Instructions

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Step 9: From the Execute Business Process screen, view the result. For more information on the executed business process click info under the Instance Data column. The screen shot shows the sample result.

Execute Business Process

Name: 6F86Gmailbox_add Instance ID: 7504 User: abcde

Completed
Status: Success

Step	Service	Status	Advanced Status	Started	Ended	Status Report	Document	Instance Data
1	Mailbox Add Service	Success	None	03/23/2016 5:52:44 AM EDT	03/23/2016 5:52:44 AM EDT	Info	Info	Info
0	INITIATING_CONTEXT from user 'fg_sysadmin'	Success	None	03/23/2016 5:52:44 AM EDT	03/23/2016 5:52:44 AM EDT	None	Info	Info

* Inline Invocation

Instance Data

This page enables you to view process data, the message to the service, and the message from the service that have been processed.

- Click **Process Data** to view the process data.
- Click **Message To Service** to view the message to the service.
- Click **Message From Service** to view the message from the service.

[Process Data](#)
[Message To Service](#)
[Message From Service](#)

Process Name: 6F86Gmailbox_add Instance ID: 7504
Service Name: MailboxAdd

```
<?xml version="1.0" encoding="UTF-8"?>
<ProcessData>
  <PrimaryDocument SCIObjctID="256276153a2dd016fnode1"/>
  <TRACKINGID>666286153a2dd016fnode1</TRACKINGID>
  <MessageId>10</MessageId>
  <DocumentId>546274153a2dd016fnode1</DocumentId>
  <CreateDateTime>2016-03-23 05:52:44.794</CreateDateTime>
  <MessageName>partner_NewFile.txt</MessageName>
  <MailboxPath>/</MailboxPath>
  <ContentType>application/data</ContentType>
  <MessageSize>38</MessageSize>
  <ExtractableCount>1</ExtractableCount>
</ProcessData>
```

Step 10: Click **SCIObjctID** hyperlink to view the primary document of **partner_NewFile.txt**.

Step 11: Close the window and log out from Sterling B2B Integrator.

Analyze the result

Complete the following steps to analyze the routed file in filegateway.

Step 1: Log into **filegateway** by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Click **Routes > Activity**.

Step 3: Click **Find**.

Step 4: Click **Arrived Files** tab to view the routed files.

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Implementation

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Exercise 2.1.2: Build mailbox add business process

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Instructions

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Complete the following steps to analyze the files in partner user interface.

Step 1: Log into myfilegateway by specifying **partner** as the user ID and **password** as password.

Step 2: Select the **File Activity** tab.

Step 3: Click **Find** button in Search Criteria panel to view the arrived files in abcde's inbox.

Step 4: Click **Logout**.



Note: _____

You can check the layer of the .zip file. The format of the file container should be as stated in the consumer facts of the routing channel template.

Business process decides consumer

Consumer identification policy

The consumer identification policy is the method Sterling File Gateway uses to identify the consumer to receive the file transfer. The consumer identification policy is defined in the routing channel template.

To implement a specific policy:

1. Write a business process to implement the consumer identification policy. For example, your business process can contain a map to derive the consumer name. Note the element name in ProcessData that contains the consumer name.
 2. When you create a routing channel template, select Dynamic for Template Type.
 3. Under *What will determine the consumer?*, select Business Process.
 4. Specify the business process name.
 5. Specify the element name in ProcessData that contains the consumer name.
-

Exercise 2.1.3: Create inbound dynamic route business process

Introduction

In this exercise you create a business process that determines the consumer's mailbox based on a database query.

The exercise is a five step process:

1. Create a code list table in Sterling B2B Integrator to map a producer/consumer relationship.
2. Review the business process that SFG invokes. SFG then determines the consumer's mailbox.
3. Create a Dynamic Inbound Routing Channel Template.
4. Create a Routing Channel.
5. Upload the file and analyze the result.

Instructions

Create a code list

Complete the following steps to create a code list:

Step 1: Log in to **filegateway** by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: Click **Trading Partner > Code Lists**.

Step 4: From the create **New Code List**, select **Go!**

Step 5: Enter the Code List name as **SFG** and click **Next**.

Step 6: Click **+add New Code**.

Sender Code	Receive Code	Description
partner	abcde	SFG

Step 7: Click **Save**.

Step 8: Click **Next** and click **Finish**.

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Exercise 2.1.3: Create inbound dynamic route business process

(Continued)

Instructions

Review the Consumers Mailbox Business Process:

Complete the following steps to review the consumer's mailbox business process:

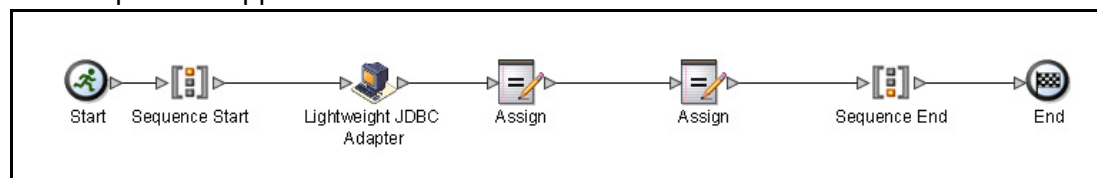
Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: Open the **GPM**.

Step 4: From GPM tool, click **File > Open**.

Step 5: Access **6F86G_consumer_name.bp** file from **C:\6F86G\ClassFiles** directory. The business process appears as shown in the screen shot:



Step 6: Click the Lightweight JDBC Adapter stencil. The Lightweight JDBC Adapter performs an SQL select statement using the XPath expression of `"/ProcessData/ProducerName/text()`. SFG populates this tag. The adapter looks up the producer's name in the code list and returns the consumer's name in a tag called `"RECEIVER_ITEM"`.

Step 7: Click the first Assign stencil. The assign service needs to move the primary document into process data. The Lightweight JDBC adapter always creates a primary document in XML format. Once in process data, the business process can use the database query to route the data.

Step 8: Click the second Assign stencil. The assign service sets a tag (ConsumerName) in process data by converting the tag to `RECEIVER_ITEM`. The Lightweight JDBC Adapter returns the `RECEIVER_ITEM`.

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Exercise 2.1.3: Create inbound dynamic route business process

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Instructions

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Check in the business process

Complete the following steps to check in the business process:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: From the **B2B Console**, click **Business Process > Manager**.

Step 4: From the **Create** section, next to **Process Definition**, click **Go!**

Step 5: From the Editor: **Process Name** dialog box, specify the following values:

Parameter	Value
Name	6F86G_consumer_name
Select an input mode for defining the new process	Check in business process created by the graphical modeling tools

Step 6: Click **Next**.

Step 7: From the Editor: Process: **6F86G_consumer_name**: Check-in dialog box, click **Browse** to open **6F86Gconsumer_name.bp** from C:\6F86G\ClassFiles folder.

Step 8: From the Editor: Process: **6F86G_consumer_name**: Check-in dialog box, in the Description text box, type **Mailbox Database Inquiry** and click **Next**.

Step 9: Accept the default values for the remaining parameters. Finally, verify that the Business Process is enabled. Click **Finish**.

Create dynamic routing channel template

Complete the following steps to create dynamic routing channel template:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: From the main menu, click **Routes > Templates**.

Step 3: Click **+Create**.

Step 4: Enter Template Name: **Dynamic_inbound**.

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Exercise 2.1.3: Create inbound dynamic route business process

(Continued)

Instructions

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Step 5: Enter Template Type: **Dynamic**. Specify the following values:

Parameter	Value
What will determine the consumer?	Business Process
Business process Name	6F86G_consumer_name
ProcessData Element Name	ConsumerName

Step 6: Click **Special Characters** tab. Select **None** for the **Special Characters**.

Step 7: Click **Groups** tab. Click, **+Add** to add the **Producer Groups**.

Step 8: From the drop down list, select **All Partners**.

Step 9: Click, **+Add** to add the **Consumers Groups**.

Step 10: From the drop down list, select **All Partners**.

Step 11: Click the **Provisioning Facts** tab. Accept the default values for the **Provisioning Facts**.

Step 12: Click the **Producer** tab. Click, **+Add** to add the **Producer**.

Step 13: In the **Producer File Structure** window, specify the following values:

Parameter	Values
Producer File Type:	Text
File name pattern as regular expression	(.+).txt
File name pattern group fact names, comma delimited:	myFilename

Step 14: Click **Save**.

Step 15: Click **Consumer** tab.

Step 16: Click **+Add** to add the **Consumer**.

Step 17: Accept the default values for mailbox path.

Step 18: Select **If checked, mailboxes matching this pattern may be created on demand**.

Step 19: Click **+Add**.

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Exercise 2.1.3: Create inbound dynamic route business process

(Continued)

Instructions

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Step 20: In the **Consumer File Structure** window, specify the following values:

Parameter	Value
Consumer File Type:	Text
File name format:	\${myFilename}.txt

Step 21: Click **Save** three times. The **Routing Channel Template successfully created** message is displayed. Click **OK**.



Note: _____

Observe the routing channel template summary windows for the detailed configuration.

Create routing channel

Complete the following steps to create routing channel:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: From the main menu, click **Routes > Channels**.

Step 3: Click **+Create**.

Step 4: From the **Create a New Routing Channel** window, specify the following values:

Parameter	Value
Routing Channel Template:	Dynamic_inbound
Producer:	partner

Step 5: Click **Save**. The system replies, **Successfully created Routing Channel** for: select **OK**.

Step 6: Log out of filegateway.

Exercise 2.1.3: Create inbound dynamic route business process

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Instructions

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Upload input data into myfilegateway

Complete the following steps to upload input data into myfilegateway:

Step 1: Log in to **myfilegateway** by specifying **partner** as user ID and **password** as password.

Step 2: Click **Upload Files** tab.

Step 3: From the Mailbox path section, select, / [forward slash].

Step 4: From the File section, select **C:\6F86G\ClassFiles\NewFile.txt**.

Step 5: Click **Send**. The file is routed to /abcde/inbox.

Step 6: Open *http://192.168.40.100:9000/mailbox* in a web browser.

The screenshot shows the IBM Sterling B2B Integrator web interface. At the top is a blue header with the IBM logo and 'Sterling B2B Integrator' text. Below the header is a navigation bar with a blue button labeled 'Mailbox Search'. To the left of the main content area is a vertical menu with links: 'Admin Console Home', 'Mailbox Search', 'Mailbox Send', 'Change Password', 'Log Out', 'Refresh Mailboxes', and 'Help'. The main content area is titled 'Mailbox Search' and contains a 'Search Options' section. This section includes a 'Mailbox' field with the value '/abcde/inbox', a 'MailboxPath lookup' link, and input fields for 'Message Name' and 'Message ID'. Below these is a 'Creation Date' section with 'From' and 'To' date and time pickers. The 'From' picker shows '2016-03-22 7:11 AM' and the 'To' picker shows '2016-03-23 7:12 AM'. A 'Go!' button is at the bottom right of the search options.

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Exercise 2.1.3: Create inbound dynamic route business process

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Instructions

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Step 7: Click **Go!** to view the mailbox result of /abcde/inbox.

Mailbox Search Results

Welcome fg_sysadmin

Extract	Message Name	Id	Created	Size	Mailbox	Extract Policy	Policy Value
	InventoryOf3.xml	7	2016-03-23 00:06:44.141	232	/abcde/inbox	Count	1
	NewFile.txt	17	2016-03-23 07:00:49.907	38	/abcde/inbox	Count	1
	NewFile.txt	16	2016-03-23 07:00:44.964	38	/abcde/inbox	Count	1
	NewFile.txt	19	2016-03-23 07:01:07.037	38	/abcde/inbox	Count	1
	Safety.GIF	5	2016-03-22 07:33:31.65	70102	/abcde/inbox	Count	0

Items 1 - 5 of 5
Items per page: [5](#) [10](#) [25](#) [50](#) [100](#)

Last update time: Wed Mar 23 2016 04:12:32 GMT-0700 (Pacific Standard Time)

Step 8: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 9: Click **Routes > Activity**.

Step 10: Click **Find** to verify the process completed successfully as shown in the screen shot.

IBM Sterling File Gateway

Welcome fg_sysadmin

Routes Participants Tools Profile Help Sign Out

Search Criteria Arrived File

Times shown in local time zone [Change to server time zone](#)

The number of search results found: 10

Reviewed	Arrived Fil...	Delivery Status	Producer	Original File Name	Discovery Time
☐	☐	Routed	Complete	partner	NewFile.txt
					03/23/2016 04:01:06

Broadcast Messages

Introduction

Following are the attributes of broadcast messages:

- You can create dynamic routing channel templates to route copies of the same file to multiple consumers.
 - The set of consumers receiving a copy of the payload file can be defined either through the system fact *ConsumerBroadcastGroup* or through a business process that returns multiple consumer partner names.
 - You can configure multi-consumer broadcast message during Routing Channel Template, Routing Channel, and Shared Mailboxes.
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Exercise 2.1.4: Broadcast message through dynamic routing

Introduction

In this exercise, the producer uploads consumer.txt file, which is broadcast to consumer.

The exercise of following steps:

1. Create a consumer group and assign a partner to receive message.
2. Create a producer partner and assign a partner to produce message.
3. Associate producer partner to producer group
4. Create routing channel template to broadcast the message.
5. Create a routing channel.
6. Broadcast a message.
7. Analyze the result in filegateway, Sterling B2B Integrator and Mailbox.

Instructions

Create consumer group and assign a partner to receive message

Complete the following steps to create consumer group and assign a partner:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the main menu, click **Participants > Groups**.

Step 3: Click **+Create**.

Step 4: From the Create a New Group screen, specify the following value:

Parameter	Value
Partner Group Name	Consumer

Step 5: Click **Save** and click **OK**.

Step 6: Select the Consumer group from the Sterling File Gateway Groups section.

Step 7: Click **Add Partners** below the Partners that belong to the Selected Group section.

Step 8: From Add partners, Add Identities to Groups screen, select **abcde** and **partner** from Identities section.

Step 9: Select **Consumer** group from the Groups section and click **Execute**. In the confirmation window.

Step 10: Click **OK** to close the window.

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Exercise 2.1.4: Broadcast message through dynamic routing

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Instructions

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Create producer group

Complete the following steps to create producer group:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: From the main menu, click **Participants > Groups**.

Step 3: Click **+Create**.

Step 4: In the Create a New Group pop-up window, specify the following value:

Field	Parameter
Partner Group Name	Producer

Step 5: Click, **Save**. The system displays the message **Created Group: Producer**. Click **OK**.

Complete the following steps to create broadcast partner:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: From the main menu, click **Participants > Partners**.

Step 3: Click **+Create**.

Step 4: In the Add Partner screen, select your Community from the drop down list, **SFG**.

Step 5: Click **Next**.

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Exercise 2.1.4: Broadcast message through dynamic routing

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Instructions

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Step 6: In the Add Partner: Information screen, specify the following values

Parameter	Value
Partner Name:	broadcastpartner
Partner Code:	<leave blank>
Address:	123 E Street
City:	Ann South Arbor
State:	MI
Postal Code:	12345
Phone:	123-456-1234
Country:	UNITED STATES
Time Zone:	(GMT-05:00) Eastern Time (US & Canada)
Email Address:	student1@ibm.com

Step 7: Click **Next**.

Step 8: In the **Add Partner: User Account** screen, specify the following values:

Parameter	Value
Authentication Type:	Local (Note: External is for LDAP password control)
User name:	broadcastpartner
Password:	password
Confirm Password:	password
Password Policy:	<leave blank>
Authentication Host:	<leave blank>
Session Timeout:	30
Given Name:	broadcastpartner
Surname:	broadcastpartner

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Exercise 2.1.4: Broadcast message through dynamic routing

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instructions

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Step 9: Click **Next**.

Step 10: In the **Add Partner: Partner Role** screen, specify the following values:

Parameter	Value
Partner is a Consumer of Data:	Leave blank
Partner will initiate the connection to consume data:	Leave blank
Partner is a Producer of Data:	Check

Step 11: Click **Next**.

Step 12: In the **Add Partner: Initiate Connection Settings** screen, specify the following value:

Parameter	Value
Will partner use either SSH/SFTP protocol to initiate connections?	No

Step 13: Click **Next**.

Step 14: In the **PGP Settings** screen, click **Next**.

Step 15: Click **Finish**. The system displays the message **Added Partner Successful**” Click **Return** and close the window.

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

Assign a partner to receive message

Complete the following steps to assign a partner to receive message:

- Step 1:** Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.
- Step 2:** From main menu, click **Participants > Groups**.
- Step 3:** Select the **Producer** group from the Sterling File Gateway Groups section.
- Step 4:** From the Selected Group panel, click **Add Partners**.
- Step 5:** From the **Add Partners, Add Identities to Groups**, select **broadcastpartner** from Identities section.
- Step 6:** Select the **Producer** group from the Groups section and select **Execute**. The following message displays the confirmation screen.



- Step 7:** Review the information in the confirmation screen and click **OK**.
- Step 8:** Close the window.

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

Create a routing channel template to broadcast message

Complete the following steps to create a routing channel template:

Step 9: Log in to Sterling File Gateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 10: Click **Routes > Templates**.

Step 11: Click **+Create**.

Step 12: In the **Template Name** field, enter **Training_Broadcast_RCT**.

Step 13: From the **Type** tab, select **Dynamic**.

Step 14: From the **What will determine the consumer** section, select **Facts in producer file structure**.

Step 15: Enter the partner group details.

- Click the **Groups** tab.
- Click **+Add** under the **Producer Groups** section.
- In the Producer Groups drop down list, select **Producer**.
- Click **+Add** under the **Consumer Groups** section.
- In the **Consumer Groups** drop down list, select **Consumer**.

Step 16: Click Producer tab.

Step 17: Enter the producer details.

- Click the **Producer** tab.
- In the **Pattern for Producer Mailbox Path** field, enter **/\${ProducerName}/Broadcast_nonShared**.
- Click **+Add** under the **Producer File Structure Description** section.
- From the **Producer File Type** drop down list, select **Text**.
- From the **File name pattern** field, enter **(.+).txt**.
- From the **File name pattern group fact names** field, enter **ConsumerBroadcastGroup**.
- Click **Save**.

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

Step 18: Enter the consumer details.

- Click the **Consumer** tab.
- From the **Delivery Channel Description** section, Click **+Add**.
- In the **Pattern for Consumer Mailbox Path** parameter, specify **/Broadcast_Inbox**.
- In the Consumer File Structure window, click **+Add**.
- In the **Consumer File Type** drop down list, select **Text**.
- In the **File Name Format** field, enter **\${ProducerName}To_\${ConsumerName}.txt**.
- Click **Save**.
- Click **Save** in the New Delivery Channel window.

Step 19: Click **Save**. The system displays routing channel created successfully. Click **OK**.

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

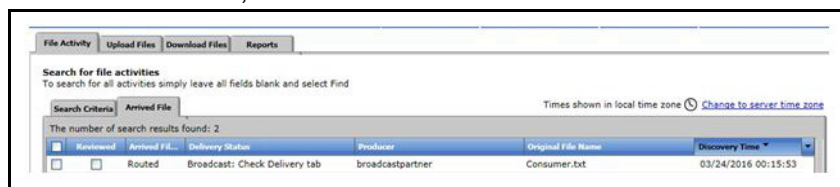
Create a routing channel

Complete the following steps to create a routing channel:

- Step 1:** Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.
- Step 2:** Click **Routes > Channels**. The list of existing routing channels is shown.
- Step 3:** Click **+Create**.
- Step 4:** From the **Routing Channel Template** drop down list, select **Training_Broadcast_RCT** template.
- Step 5:** From the **Producer** drop down list, select **broadcastpartner**.
- Step 6:** Click **Save**. A message stating outgoing channel is added to the list is displayed. Click **OK**.

Broadcast messages

- Step 1:** In the windows image, access **C:\6F86G\ClassFiles** folder, rename **consumer.txt** to **Consumer.txt**.
- Step 2:** Log in to myfilegateway by specifying **broadcastpartner** as the user ID and **password** as password.
- Step 3:** Click the **Upload files** tab.
- Step 4:** From the **Mailbox Path** drop down list, select the **Broadcast_nonShared**.
- Step 5:** Browse and select the **Consumer.txt** file from the 6F86G\ClassFiles folder.
- Step 6:** Click **Send**. The file is routed, as shown in the screen shot.



Known	Arrived File	Delivery Station	Producer	Original File Name	Discovery Time
☐	Routed	Broadcast: Check Delivery tab	broadcastpartner	Consumer.txt	03/24/2016 00:15:53

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

Analyze the result in filegateway

Complete the following steps to analyze the result:

Step 1: Log into filegateway <http://192.168.40.100:9000/filegateway> with specifying ID **fg_sysadmin** as user ID and **password** as password.

Step 2: Click **Routes > Activity**.

Step 3: Click **Find**.



Step 4: Click Routed record to view the details of transferred file. Make a note of the process workflow ID.

For example, if the workflow ID is 8317, select the workflow ID 8317 in Sterling B2B Integrator to view the successful execution of the business process.

Follow the steps to view the workflow

Complete the following steps to view the workflow:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Click **Tools > B2B Console**.

Step 3: From Sterling B2B Integrator interface, login with **admin** as user ID, **password** as password.

Step 4: From Admin Console, navigate to **Business Process > Monitor > Advanced Search > Business Processes**.

Step 5: Enter **Process ID:** as 8317 and Click **Go!**.

Step 6: Click Instance ID hyper link. and navigate to the page 2 to view details.

(Continued on next page)

Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

....(Continued)

Step 7: In the Business Process Detail page, locate the File Gateway Delivery Service, click icon in the Status Report column to view the details of the business process. The following screen shot displays the status of the business process where the file is delivered to the /Broadcast_Inbox folder.

Step 8: In the Business Process Detail page, locate the FileGatewayRoutedArrivedFile Service, click icon in the Status Report column to view the details of the business process. The following screen shot displays the status of the business process where the file is delivered to the /Broadcast_Inbox folder.

Step 9: The following screen shot highlights the details of the file name, mailbox path, and layer.

(Continued on next page)

Implementation

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Exercise 2.1.4: Broadcast message through dynamic routing

(Continued)

Instructions

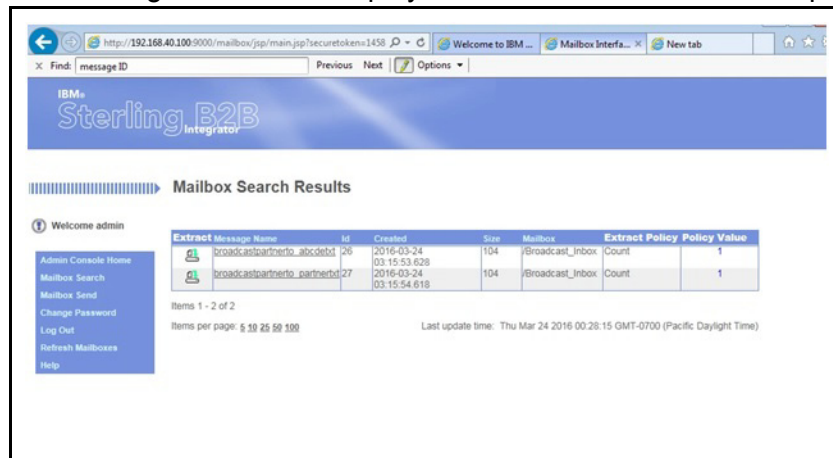
....(Continued)

Complete the following steps to analyze the file transfer in the mailbox:

Step 1: From a web browser, open <http://192.168.40.100:9000/mailbox>.

Step 2: From the menu, select Mailbox Search and type /Broadcast_Inbox in the Search Option filed

Step 3: Click **Go!** The following screen shot displays the Mailbox Search Results page.



Shared and linked mailboxes

Introduction

Linking trading partner mailboxes to shared mailboxes allows trading partners to view real-time data that is stored in the shared mailboxes. The linked mailbox has read-only copy of the shared mailbox and the trading partners cannot modify or delete data in it.

To enable the shared and linked mailbox feature in Sterling B2B Integrator, set the following property in the `customer_overrides.properties` file as:

`mailbox.enableSharedLinkedMailboxes=true`

By default, this property is set to *false*.

This property change adds an additional field *Mailbox Type* to the mailbox configuration page.

Sterling File Gateway creates new or reuses existing mailboxes as necessary to support producer drop of and consumer pickup of messages for a routing channel.

Exercise 2.1.5: Broadcast messages using shared mailboxes

Introduction

In this exercise, you will broadcast messages through shared mailbox.

The exercise is a four step process:

1. Create a partner.
2. Create a routing channel template to broadcast.
3. Create a routing channel.
4. Create a linked mailbox and broadcast a message.

Instructions

Create a partner

Complete the following steps to create broadcast partner.

Step 1: Log in to filegateway by specifying **fg_sysadmin** as user ID and **password** as password.

Step 2: From the main menu, navigate to **Participants > Partners**.

Step 3: Click **+Create**.

Step 4: From the Add Partner screen, select your Community from the drop down list, **Producer**, and click **Next**.

Step 5: In the Add Partner: Information screen, specify the following values:

Parameter	Value
Partner Name:	SharedMB
Partner Code:	<leave blank>
Address:	123 E Street
City:	Ann South Arbor
State:	MI
Postal Code:	12345
Phone:	123-456-1234
Country:	UNITED STATES
Time Zone:	(GMT-05:00) Eastern Time (US & Canada)
Email Address:	student1@ibm.com

Step 6: Click Next.

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Step 7: In the **Add Partner: User Account** screen, specify the following values:

Parameter	Value
Authentication Type:	Local (Note: External is for LDAP password control)
User name:	broadcastpartner
Password:	password
Confirm Password:	password
Password Policy:	<leave blank>
Authentication Host:	<leave blank>
Session Timeout:	30
Given Name:	broadcastpartner
Surname:	broadcastpartner

Step 8: Click **Next**.

Step 9: In the **Add Partner: Partner Role** screen, specify the following values:

Parameter	Value
Partner is a Consumer of Data:	Leave blank
Partner will initiate the connection to consume data:	Leave blank
Partner is a Producer of Data:	Check

Step 10: Click **Next**.

Step 11: In the **Add Partner: Initiate Connection Settings** screen, specify the following value:

Parameter	Value
Will partner use either SSH/SFTP protocol to initiate connections?	No

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Step 12: Click **Next**.

Step 13: In the **PGP Settings** screen and click **Next**.

Step 14: Click **Finish**. The system displays the message **Added Partner Successful**. Click **Return** and close the window.

Assign partner to the producer group

Complete the following steps to assign the partner to the producer group:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the main menu navigate to **Participants > Groups**.

Step 3: Select the **Producer** group from the Sterling File Gateway Groups section.

Step 4: In the **Add Partners, Add Identities to Groups** window, click **Add Partners** in the select group section.

Step 5: Select **SharedMB** from Identities section.

Step 6: Select **Producer** group from the Groups section and click **Execute**.

Step 7: Review the information in the confirmation screen and select **OK**.

Step 8: Close the window.

(Continued on next page)

Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Create a routing channel template to broadcast message

Complete the following steps to create a routing channel template:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Navigate to **Routes >> Templates**.

Step 3: Click **+Create**.

Step 4: In the **Template Name** field, enter **Training_RCT**.

Step 5: From the **Type** tab, select **Dynamic**.

Step 6: From the **What will determine the consumer** section, select **Facts in producer file structure**.

Step 7: Specify the partner group details.

- Click the **Groups** tab.
- Click **+Add** under the **Producer Groups** section.
- From the **Producer Groups** drop down list, select **Producer** group.
- Click **+Add** under the **Consumer Groups** section.
- From the **Consumer Groups** drop down list, select **Consumer** group.

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Step 8: Specify the producer details.

- Click the **Producer** tab.
- In the **Pattern for Producer Mailbox Path** field, enter **/\${ProducerName}/Shared_Prod_MB**.
- Click **+Add** under the **Producer File Structure Description** section. The Producer File Structure window opens.
- From the **Producer File Type** dropdown list, select **Text**.
- In the **File name pattern** field, enter **(.+).txt**.
- In the **File name pattern group fact names** field, enter **ConsumerBroadcastGroup**.
- Click **Save**.

Step 9: Specify the consumer details.

- Click the **Consumer** tab.
- Click **+Add** under the **Delivery Channel Description** section. The New Delivery Channel window opens.
- In the **Pattern for Consumer Mailbox Path** field, enter **/Shared_MB**. This mailbox is the shared mailbox that is created in Sterling B2B Integrator.
- Click **+Add** to create the consumer file structure. The Consumer File Structure window opens.
- From the **Consumer File Type** dropdown list, select **Text**.
- In the **File Name Format** field, enter **\${ProducerName}To_\${ConsumerName}.txt**.
- Click **Save** twice.

Step 10: Click **Save** in the routing channel template screen.

Step 11: Click **OK**.

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Create a routing channel

Complete the following steps to create a routing channel:

- Step 1:** Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.
- Step 2:** Navigate to **Routes > Channels**.
- Step 3:** Click **+Create**.
- Step 4:** From the **Routing Channel Template** drop down list, select the **Training_RCT** template.
- Step 5:** From the **Producer** drop down list, select **SharedMB**.
- Step 6:** Click **Save**.
- Step 7:** Click **OK**. The routing channel is added to the list.

Create a Linked Mailbox

Complete the following steps to create a linked mailbox:

- Step 1:** Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.
- Step 2:** Navigate to **Tools > B2B Console**.
- Step 3:** Log in to Sterling B2B Integrator by specifying **fg_sysadmin** as the user ID and **password** as password.
- Step 4:** From the **Administration Menu**, Navigate to **Deployment > Mailboxes > Configuration**.
- Step 5:** Click **Create** and name the mailbox as **Linked_Cons_MB**.
- Step 6:** Move **SharedMB** and **fg_sysadmin** users from **Available users** to **Selected Users**.
- Step 7:** Click **Next**.
- Step 8:** Click **Save** and click **Finish**.

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

Instructions

....(Continued)

Broadcast messages using Shared Mailbox

Complete the following steps to broadcast messages using shared mailbox:

Step 1: Log in to myfilegateway by specifying **SharedMB** as the user ID and **password** as password.

Step 2: Click **Upload files** tab.

Step 3: From the **Mailbox Path** drop down list, select the shared mailbox, **Shared_Prod_MB**.

Step 4: Browse and select the **consumer.txt** file from the **C:\6F86G\ClassFiles** folder.

Step 5: Click **Send**, to view the routed file, as shown in the screen shot.

The screenshot shows the IBM Sterling myFileGateway web interface. At the top, there's a header with 'IBM Sterling myFileGateway' and 'Welcome SharedMB'. Below the header are navigation tabs: 'Home', 'Profile', and 'Help'. Underneath, there are more tabs: 'File Activity', 'Upload Files', 'Download Files', and 'Reports'. The 'File Activity' tab is selected. Below the tabs, there's a search section titled 'Search for file activities' with a subtext 'To search for all activities simply leave all fields blank and select Find'. There are two sub-tabs: 'Search Criteria' and 'Arrived File'. The 'Arrived File' tab is selected. Below the search section, it says 'The number of search results found: 1'. A table displays the search results with columns: 'Reviewed', 'Arrived Fil...', 'Delivery Status', 'Producer', 'Original File Name', and 'Discover'. The table contains one row with the following data: 'Routed', 'Broadcast: Check Delivery tab', 'SharedMB', 'Consumer.txt', and '03/24/2'.

Reviewed	Arrived Fil...	Delivery Status	Producer	Original File Name	Discover
☐	☐	Routed	Broadcast: Check Delivery tab	SharedMB	Consumer.txt

Step 6: Log out of myfilegateway.

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Exercise 2.1.5: Broadcast messages using shared mailboxes

(Continued)

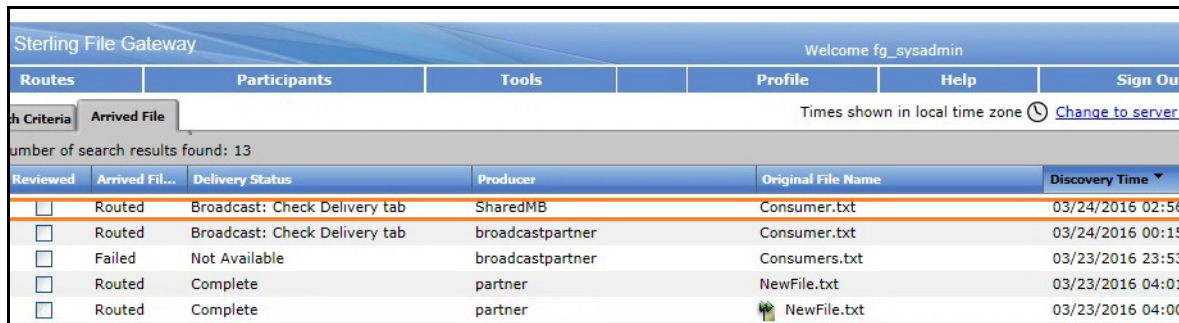
Instructions

....(Continued)

Step 7: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password..

Step 8: Select **Route > Activity**.

Step 9: Click **Find** to view the lists of routed files.



The screenshot shows the 'Sterling File Gateway' interface with the 'Arrived File' tab selected. The table displays search results for routed files. The first row is highlighted in orange.

Reviewed	Arrived Fil...	Delivery Status	Producer	Original File Name	Discovery Time
☐	Routed	Broadcast: Check Delivery tab	SharedMB	Consumer.txt	03/24/2016 02:56
☐	Routed	Broadcast: Check Delivery tab	broadcastpartner	Consumer.txt	03/24/2016 00:15
☐	Failed	Not Available	broadcastpartner	Consumers.txt	03/23/2016 23:53
☐	Routed	Complete	partner	NewFile.txt	03/23/2016 04:03
☐	Routed	Complete	partner	NewFile.txt	03/23/2016 04:00

Step 10: Double click the latest routed file of SharedMB producer to analyze the information.

Lesson review

What you have been able to do

- Describe the major functionality of dynamic routing channel.
 - Configure a dynamic routing channel template.
 - Apply business process to determine consumers.
 - Explain and configure broadcast messages using dynamic routing channel template.
 - Explain mass replay and redelivery of processed transaction.
-